Three-Currency Economics on Holochain: SAP, TEND, and MYCEL with Consciousness-Gated Treasury

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Abstract

We present a three-currency economic system implemented on Holochain's agent-centric architecture with 8 coordinator zomes. The system distinguishes three economic functions—stable accounting (SAP), care-oriented mutual credit (TEND), and commons stewardship (MYCEL)—each with distinct monetary properties: SAP is commodity-backed and demurrage-bearing; TEND is bounded zero-sum mutual credit without demurrage; MYCEL represents collective stewardship and funds public goods. Treasury operations are consciousness-gated, requiring progressive governance tiers for increasing levels of financial authority. We describe the implementation of four critical financial operations—oracle price feeds, mutual credit issuance, concurrent demurrage computation, and currency composting—and report on hardening fixes for division-by-zero, TEND race conditions, concurrent demurrage overflow, and oracle-to-TEND call ordering. All 10 financial operations are consciousness-gated through shared verification functions, and the system includes Sybil resistance through consciousness-tier requirements on currency minting.

1 Introduction

Most cryptocurrency projects implement a single currency optimized for a single function: Bitcoin for store of value, stablecoins for unit of account, Ethereum for programmable transactions. Real economies, however, require multiple monetary instruments serving distinct purposes [1].

Bernard Lietaer argued that monetary monocultures are fragile: when a single currency fails, the entire economy suffers. Complementary currencies—local exchange systems, time banks, mutual credit networks—provide resilience through diversity [2].

We implement this insight on Holochain with three currencies, each designed for a different economic function:

SAP (Sapience) Stable unit of account. Backed by a basket of real goods and services. Used for pricing, contracts, and inter-community exchange.

TEND (Tenderness) Mutual credit for care work and community service. Implements demurrage (value decay) to encourage circulation. Named for the tenderness of care labor.

MYCEL (Mycelium) Commons currency for collective stewardship. Used for treasury operations, public goods funding, and regenerative investment.

2 Architecture

2.1 Eight Zomes

1. **currency-mint**: Supply management for all three currencies. Minting requires consciousness-gated authorization.
2. **payments**: Transaction processing. Handles transfers, batch payments, and cross-

currency exchanges.

3. **tend**: TEND-specific mutual credit operations including credit line management, demurrage computation, and composting.
4. **treasury**: Commons treasury management with multi-signature (DKG) authorization for large expenditures.
5. **staking**: MYCEL staking for governance weight amplification.
6. **recognition**: Non-monetary recognition tokens for gratitude and appreciation.
7. **price-oracle**: Decentralized price feeds for the SAP basket.
8. **bridge**: Cross-cluster financial dispatch.

2.2 Holochain Advantages for Finance

Holochain's agent-centric model provides several advantages for financial applications:

- **Double-spend prevention**: Each agent's source chain is append-only. A double-spend would require forking the source chain, which is detectable by DHT validators.
- **No transaction fees**: Unlike blockchain, Holochain has no per-transaction gas fees. This makes micro-transactions viable.
- **Offline transactions**: Agents can create transactions on their source chain while offline, syncing when connectivity returns.
- **Privacy**: Balances are on the agent's source chain, not a public ledger. Only the counterparty and DHT validators see transaction details.

3 SAP: Stable Unit of Account

3.1 Basket Backing

SAP is pegged to a basket of real goods and services, defined by the community. A typical basket might include:

- 1 kg of locally-grown grain
- 1 hour of basic skilled labor
- 1 kWh of renewable energy
- 1 liter of clean water

The basket composition is governed by the Governance cluster (Steward+ required to modify). The price-oracle zome aggregates price attestations for each basket component.

3.2 Minting

SAP cannot be created ex nihilo. Minting occurs only when:

1. An agent contributes a basket-equivalent of real goods/services to the community
2. The contribution is verified by at least two Citizen-tier attestors
3. The currency-mint zome issues SAP proportional to the basket value

This makes SAP supply a function of real economic activity rather than arbitrary monetary policy.

4 TEND: Mutual Credit Without Demurrage

4.1 Mutual Credit

TEND operates as a mutual credit system [4]: every TEND created as a positive balance simultaneously creates an equal negative balance. The total TEND supply across the network is always zero:

$$\sum_i B_i^{\text{TEND}}(t) = 0 \quad \forall t \tag{1}$$

This property is enforced by the integrity zome: the validation rule verifies that each TEND transaction creates equal and opposite entries on the sender's and receiver's source chains.

4.2 Balance Discipline

TEND does not decay over time. Instead, circulation discipline is enforced by reciprocal accounting and bounded credit lines. Each member's positive or negative position is constrained by consciousness-gated limits, and the entire system remains zero-sum:

$$\sum_i B_i^{\text{TEND}}(t) = 0 \quad \forall t \tag{2}$$

Proposition 1 (TEND zero-sum invariant)**.** *For every confirmed exchange, the provider's positive balance increase exactly matches the receiver's negative balance increase:*

$$\Delta B_{provider} + \Delta B_{receiver} = 0 \tag{3}$$

Proof. Each demurrage deduction from a positive balance is added to the compost treasury. Each demurrage reduction of a negative balance is subtracted from the compost treasury. Both operations preserve the zero-sum invariant. □

4.3 Concurrent Demurrage

A critical implementation challenge: demurrage must be computed correctly when multiple transactions interact with the same balance concurrently.

The initial implementation had a counter overflow vulnerability: if two concurrent transactions each computed demurrage on the same balance, the second transaction would double-count the decay. The fix: demurrage is computed exactly once per balance-modifying operation, using the source chain's timestamp ordering as the canonical timeline.

5 MYCEL: Commons Currency

5.1 Collective Stewardship

MYCEL represents collective stewardship capacity. It is:

- Minted by community-wide consensus (Governance cluster)
- Held in the commons treasury
- Spent on public goods, infrastructure, and regenerative investment
- Stakeable for governance weight amplification

5.2 Staking

MYCEL staking locks tokens for a configurable period, providing:

- Governance weight amplification (staked MYCEL increases vote weight)
- Treasury stability (locked tokens reduce circulating supply)
- Commitment signaling (staking demonstrates long-term alignment)

Staking does not generate yield (this is not DeFi). The benefit is governance influence, not financial return.

6 Consciousness-Gated Treasury

6.1 Tiered Access

Treasury operations are gated by consciousness tier:

Operation	Min. Tier
View balance	Observer
Propose expenditure	Citizen
Approve expenditure ($< 1\%$ treasury)	Steward
Approve expenditure ($\geq 1\%$ treasury)	Steward + DKG
Emergency fund release	Guardian + DKG
Modify SAP basket	Steward + supermajo

Table 1: Consciousness-gated treasury operations.

6.2 Shared Verification Functions

All 10 consciousness-gated financial operations use two shared verification functions in `mycelix-finance/zomes/shared`:

```
pub fn verify_participant_tier(
    cred: &ConsciousnessCredential
) -> ExternResult<()> {
    if cred.tier < ConsciousnessTier::
        Participant {
        return Err(wasm_error!("Insufficient␣
            tier"));
    }
    if cred.is_expired(now_us()?) {
        return Err(wasm_error!("Expired␣
            credential"));
    }
    Ok(())
}

pub fn verify_citizen_tier(
    cred: &ConsciousnessCredential
) -> ExternResult<()> {
    if cred.tier < ConsciousnessTier::Citizen
        {
```

```
        return Err(wasm_error!("Insufficient␣
            tier"));
    }
    if cred.is_expired(now_us()?) {
        return Err(wasm_error!("Expired␣
            credential"));
    }
    Ok(())
}
```

7 Price Oracle

7.1 Decentralized Price Feeds

The price-oracle zome aggregates price attestations from multiple oracle agents:

1. Oracle agents (Citizen+ tier) submit price attestations for SAP basket components
2. Attestations include: item, price, timestamp, confidence
3. The oracle computes the median price (robust to outliers)
4. Stale attestations (beyond TTL) are excluded

7.2 Hardening

Four critical bugs were fixed during security hardening:

1. **Oracle division-by-zero**: When no valid attestations exist, the median computation divided by zero. Fix: return `None` when attestation count is zero.
2. **TEND race condition**: Concurrent TEND issuance could create unbalanced credit lines. Fix: source chain sequencing with optimistic locking.
3. **Concurrent demurrage**: Double-counted decay on concurrent balance access. Fix: single demurrage computation per balance modification.
4. **Missing oracle→TEND call**: The TEND pricing function did not call the oracle for current SAP-TEND exchange rate. Fix: added cross-zome call with fallback to last known rate.

8 Currency Composting

8.1 Lifecycle

Currencies have a lifecycle: minting → circulation → composting. Composting is the explicit removal of currency units from circulation:

- SAP: Demurrage and explicit compost routing remove idle value from circulation and fund commons maintenance
- TEND: No demurrage; circulation is governed through zero-sum credit limits rather than timed decay
- MYCEL: Governance-authorized composting of excess supply

8.2 Idempotency

Currency composting is idempotent: composting an already-composted unit is a no-op. This was a hardening fix—the original implementation allowed double-composting, which would undercount the circulating supply.

8.3 Zero-Sum Invariant

For all three currencies, the invariant holds:

$$\text{minted} - \text{composted} = \text{circulating} \tag{4}$$

This is enforced by the integrity zome and verified by DHT validators.

9 Sybil Resistance in Finance

Financial Sybil attacks (creating fake identities to obtain multiple credit lines) are prevented by:

1. **Consciousness gating**: Currency minting requires Citizen+ tier, which requires verified identity ($I \geq 0.5$)
2. **Credit line limits**: TEND credit lines are proportional to the agent's consciousness profile score
3. **Reputation consequences**: Financial misbehavior triggers reputation slashing (0.5×), affecting all governance participation

4. **Cross-cluster visibility**: The Identity cluster's web-of-trust can detect clusters of financially-interconnected Sybil identities

10 Non-Monetary Recognition

The recognition zome provides non-monetary appreciation tokens:

- Gratitude tokens (from Hearth cluster integration)
- Skill recognition (from Education cluster integration)
- Community service badges
- Milestone acknowledgments

These tokens have no monetary value but contribute to the reputation dimension of the consciousness profile, creating a positive feedback loop between community participation and governance capability.

11 Evaluation

11.1 Security Hardening Results

The four hardening fixes (Section 7.2) were discovered through a combination of code review and sweettest integration testing. The TEND race condition required a 3-agent concurrent test scenario with a BadNonce retry mechanism in the sweettest framework.

11.2 Design Trade-offs

Property	Mycelix	DeFi
Tx fees	Zero	Gas fees
Privacy	Source chain	Public ledger
Yield	None (by design)	APY incentives
Governance	Consciousness	Token-weighted
Backing	Real goods	Algorithmic

Table 2: Mycelix Finance vs. DeFi comparison.

12 Related Work

Mutual credit on Holochain: Holochain was designed partly for mutual credit [5]. Our contribution is the three-currency design with consciousness gating.

Circles UBI [6]: Ethereum-based mutual credit with social graph trust. Similar goals but limited by blockchain costs and single-currency design.

Grassroots Economics [7]: Community currency implementations in Kenya. Validates demurrage and mutual credit in practice but uses centralized infrastructure.

13 Conclusion

Three-currency economics on Holochain provides monetary diversity without the overhead of blockchain consensus. SAP, TEND, and MYCEL serve distinct economic functions—stable accounting, care labor, and collective stewardship—with consciousness-gated treasury operations ensuring that financial power scales with demonstrated governance capability rather than accumulated wealth. The hardening of four critical financial bugs demonstrates the importance of rigorous security review in decentralized financial systems, and the zero-sum invariants provide formal guarantees of monetary integrity.

References

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[7] Ruddick, W. et al. (2015). Complementary Currencies for Sustainable Development in Kenya. *IJCCR*, 19.